

Skills Summary

Adding Two-Digit Numbers with Place Value

Use this reference while completing your worksheet or when studying.

Skill 1 — Breaking both numbers into tens and ones

Rule: A two-digit number is made of tens and ones. Split both numbers, add the tens to the tens and the ones to the ones, then put the two answers back together.

- If the ones make ten or more, trade ten ones for one ten.

Example: $26 + 32$

Step 1. Both numbers are two-digit, so I split each one by place value instead of counting on.

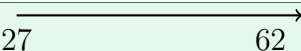
Step 2. Tens: $20 + 30 = 50$. Ones: $6 + 2 = 8$. Put them together: $50 + 8 \Rightarrow 26 + 32 = 58$

Try it: $41 + 35$

check: 92

Skill 2 — Jumping tens, then ones, on a number line

Rule: Keep the first number whole and break only the second one. Jump one ten at a time, then jump the ones. The number you land on is the total.

Example: $27 + 35$ 

Step 1. Adding on a line means I only have to break the second number, so I split 35 into three tens and five ones.

Step 2. Three ten-jumps: $27 \rightarrow 37 \rightarrow 47 \rightarrow 57$. Then a jump of five ones: $57 \rightarrow 62$.
 $\Rightarrow 27 + 35 = 62$

Try it: $32 + 24$ (two ten-jumps, then four ones)

check: 99

Skill 3 — Making a friendly ten, then adjusting

Rule: Push one number up to the nearest ten, because tens are easy to add. Then undo exactly what you added:

$$a + c = (a + b) + c - b \quad \text{where } a + b \text{ is a ten.}$$

Example: $28 + 35$

Step 1. 28 is only 2 away from 30, and adding to a ten is easy, so I lend it 2 and pay the 2 back at the end.

Step 2. $28 + 2 = 30$, then $30 + 35 = 65$, then give the 2 back: $65 - 2 = 63$. $\Rightarrow 28 + 35 = 63$

Try it: $49 + 26$ (lend 1, then pay it back)

check: 75

Skill 4 — Rounding to ten to check a sum

Rule: Look at the ones digit. If it is 5 or more, round up to the next ten; if it is 4 or less, round down. Add the rounded numbers to get an *estimate* — the real answer should land near it.

Example: About how much is $43 + 29$?

Step 1. The question says “about how much”, so I round each number to its nearest ten before adding instead of adding exactly.

Step 2. $43 \rightarrow 40$ (3 ones, round down) and $29 \rightarrow 30$ (9 ones, round up), so $40 + 30$.
 $\Rightarrow 43 + 29 \approx \mathbf{70}$

Try it: About how much is $62 + 18$?

check: 80